

Weather App

Real-Time Weather Application

Project Overview

A simple and elegant weather application that allows users to search for current weather conditions in any city worldwide. The app displays temperature, humidity, wind speed, and weather description with beautiful icons.

Technologies Used

Technology	Purpose
HTML5	Page structure
CSS3	Styling and animations
JavaScript (ES6)	Logic and interactivity
OpenWeatherMap API	Real weather data
localStorage	Search history persistence

Project Structure

text

Weather-App/

— index3.html	← Structure
— style3.css	← Styling
— script3.js	← Functionality

✨ Key Features

◆ Search Weather

- Enter city name in the input field
- Click "**Search**" button or press **Enter** key
- Displays: Temperature, humidity, wind speed, weather condition

◆ Weather Display

- **Temperature** with animated gradient
- **Weather icon** (sun, clouds, rain, etc.)
- **City name** with country
- **Description** (e.g., "clear sky", "light rain")
- **Humidity** percentage
- **Wind speed** in km/h

◆ Search History

- Automatically saves last 10 searched cities
- Click on any city in history to search again
- Clear history with "**Clear**" button

◆ Error Handling

- Shows error message for invalid city names
- Handles network errors gracefully



User Experience Walkthrough

Step	Action	Result
1	Open the app	Default weather for Amman, Jordan appears
2	Type "London"	City name appears in input field
3	Click Search or press Enter	Weather data loads with loading animation
4	Weather appears	Shows temperature, humidity, wind, and description
5	City saved to history	"London" appears in search history list
6	Click "London" from history	Weather loads again without typing
7	Click "Clear"	All search history is removed
8	Type "InvalidCity"	Error message appears "City not found"



UI Design

Colors:

- Background: Dark gradient #1a1a2e → #16213e → #0f3460
- Glass effect: rgba(255, 255, 255, 0.1) with blur
- Accent: #4fc3f7 to #2196f3 gradient
- Text: White with varying opacity

Animations:

- **Floating icon** (3-second animation)
- **Shake effect** on error
- **Hover effects** on buttons and history items
- **Smooth transitions** on all interactions

Visual Effects:

- Glass morphism (frosted glass)
- Gradient temperature text
- Subtle shadows and borders

 Example Cities to Try

City	Weather Expectation
London	Cloudy, cool
Dubai	Hot, clear sky
Tokyo	Variable
New York	Seasonal
Sydney	Sunny or rainy
Paris	Cloudy or sunny
Cairo	Hot, dry

Strengths

- ✓ **Real data** from OpenWeatherMap API
- ✓ **Beautiful design** with glass effect
- ✓ **Search history** for quick access
- ✓ **Keyboard support** (Enter key)
- ✓ **Error handling** for invalid cities
- ✓ **Responsive** on all devices
- ✓ **Auto-load** weather for default city
- ✓ **Data persistence** with localStorage

Technical Details

API Used

- **OpenWeatherMap API**
- Free API Key: bd5e378503939ddaee76f12ad7a97608
- Endpoint: <https://api.openweathermap.org/data/2.5/weather>

Data Retrieved

javascript

```
{
  name: "London",
  sys: { country: "GB" },
  main: { temp: 15, humidity: 82 },
  weather: [{ description: "clear sky" }],
  wind: { speed: 4.1 }
}
```

Icon Mapping

API Icon	Emoji
01d, 01n	☀️, 🌙
02d, 02n	☁️, ☁️
03d-04n	☁️
09d-10n	☁️, 🌧️
11d, 11n	🌧️
13d, 13n	❄️
50d, 50n	🌫️



How to Run

1. Create folder: Weather-App
2. Create files: `index.html`, `style.css`, `script.js`
3. Copy code into respective files
4. Open `index.html` in browser
5. **Internet connection required** for API calls



Future Enhancements

- **5-day forecast** feature
- **Celcius/Fahrenheit** toggle
- **Geolocation** (auto-detect user's location)
- **Dark/Light** theme toggle

- **Weather alerts** for severe conditions
- **Save favorite cities** permanently
- **Hourly forecast** display



Sample Screens

Main Screen:

- Dark gradient background with glass container
- Search bar with "Search" button
- Weather card with temperature, icon, details
- Search history below

Weather Card:

text



Amman, JO

25°C

clear sky



Humidity: 45%



Wind: 12 km/h



Quick Stats

Feature

Status

Real-time Weather



Search by City	✓
Temperature Display	✓
Humidity Display	✓
Wind Speed Display	✓
Search History	✓
Error Handling	✓
Keyboard Support	✓
Responsive Design	✓

Important Notes

- **Internet required** for API calls
- Free API key has **60 calls/minute** limit
- Use real city names (case insensitive)
- Default city loads: Amman, Jordan